

Representation Wins on QA, Not on ML: A Paired-Data Comparison of Structured FHIR and LLM-Narrative Retrieval for Adherence-Indicator Question Answering on Long-Acting Specialty Regimens

Tirthesh Jani

May 2026
arXiv preprint

Abstract

Clinical records are increasingly expressed in both structured standards (HL7 FHIR R4B) and LLM-generated narrative summaries. When both representations encode the same facts, it is unclear which should serve as the retrieval substrate for downstream question answering or adherence-risk prediction. We present a *paired-data* empirical comparison that holds information content constant: 200 synthetic patients rendered as FHIR R4B bundles and as LLM-generated narratives from the same underlying facts. Three retrieval systems answer 13,800 programmatically-verified adherence-indicator questions across five Patient Support Program (PSP) families (next-dose lookup, dose-history aggregation/MPR, coverage-window reasoning/PDC, missed-dose detection, and persistence signalling) at three regimen-complexity tiers. A fourth arm extracts features from each representation for a synthetic adherence classification task.

Results. System A (Narrative RAG) achieves 40.6% exact-match accuracy (95% CI [39.8%, 41.4%]), outperforming System B (Structured-Naive, 35.3%) and System C (Structured-Aware, 33.4%). All pairwise gaps are statistically significant (McNemar $p < 0.001$; paired bootstrap CIs exclude zero). An error taxonomy reveals that *reasoning truncation* is the dominant failure mode (44–58% of failures), disproportionately penalising the structured systems because FHIR JSON contexts require longer reasoning chains. On the feature-extraction arm, evaluated against a synthetic adherence label whose construction is partly entailed by the structured representation, FS-Structured features achieve AUC 0.997 against FS-Narrative AUC 0.843 and FS-Aware AUC 0.793; the partial tautology is examined in Results and Discussion.

Contribution. We demonstrate and empirically characterise a *representation dissociation*: the representation that wins narrative QA (narrative $>$ structured) is not the same representation that wins downstream ML adherence prediction (structured $\gg$ narrative). This is the first paired-data comparison of narrative RAG, naive structured FHIR RAG, and resource-aware structured FHIR RAG on specialty-medication adherence-indicator questions with programmatic ground truth. Code and dataset are released under Apache-2.0 / CC-BY-4.0.

1 Introduction

Specialty medications, including biologics, long-acting injectables (LAIs), and cyclic chemotherapy regimens, now account for an estimated 51% of US prescription drug spending despite representing

only a small fraction of claims [1]. In Canada, biologics alone constituted 29.6% of public drug-plan spending from just 2.4% of claims in 2022 [2]. Patient Support Programs (PSPs) provide the clinical infrastructure that tracks adherence for these high-cost, administration-intensive regimens [3]. The questions PSP case managers ask every day, such as “When is this patient’s next scheduled injection?”, “How many doses were administered in the past 90 days?”, and “Has the patient missed any doses recently?”, map directly onto canonical adherence indicators: Medication Possession Ratio (MPR), Proportion of Days Covered (PDC), and persistence [4]. Answering them requires multi-step temporal reasoning over regimen structure that standard free-text search cannot reliably perform [5].

Concurrently, LLM-powered ambient scribes have entered enterprise deployment at scale: as of 2025-2026, systems from Abridge, Nuance DAX Copilot, and Ambience Healthcare collectively serve millions of clinical encounters across academic health systems [6, 7]. These systems generate clinical narratives as a by-product of documentation, creating a practical question: when the same clinical facts are encoded in both a structured standard (FHIR) *and* an LLM-generated narrative, which representation should a retrieval-augmented generation (RAG) system [8] retrieve from?

Prior work has not answered this question because existing comparisons *confound representation with information content*. Studies on MIMIC-based predictive modelling use independently authored structured and narrative sources for the same patient, so performance differences reflect both representation and content coverage simultaneously [9]. DrugEHRQA pairs structured tables and discharge notes for medication QA but explicitly acknowledges that “information across the two modalities is not strictly disjoint” [10]. FHIR-AgentBench [11] benchmarks agent architectures over MIMIC-IV-FHIR but contains no narrative RAG arm. LLMonFHIR [12] evaluates only the structured function-calling variant. CLEAR [13] compares entity-aware vs. embedding RAG, but on narrative notes, not FHIR resources.

This paper. We resolve the confound by constructing a *paired dataset*: 200 synthetic patients each rendered as a FHIR R4B bundle and as an LLM-generated narrative derived from the same bundle, with 100% fidelity (all FHIR entities recoverable from the narrative, verified by an independent programmatic audit). Three retrieval systems answer the same 13,800 questions from both representations; differences in accuracy are therefore attributable to *representation and retrieval strategy alone*. A fourth measurement compares features derived from each representation on a synthetic adherence classification task, providing evidence about which representation is preferable for downstream ML.

Contributions.

1. **Paired-data methodology.** First head-to-head comparison of narrative RAG, naive structured FHIR RAG, and resource-aware structured FHIR RAG where information content is held constant across representations.
2. **Resource-aware FHIR retriever.** System C adds typed resource filtering, reference-chain traversal (MedicationRequest $\rightarrow$ Medication, Practitioner, Patient), and temporal pre-filtering to standard dense retrieval, extending prior schema-aware retrieval approaches [14] to the FHIR R4B resource graph.
3. **PSP adherence-indicator question grounding.** Five question families (next-dose lookup, dose-history aggregation / MPR, coverage-window / PDC, missed-dose detection, persistence) with programmatic ground truth over FHIR bundles; no LLM-as-judge.

4. **Representation dissociation finding.** The representation that wins natural-language QA (narrative > structured) is not the same one that wins downstream adherence prediction (structured $\gg$ narrative), demonstrated empirically on paired data.
5. **Error taxonomy.** Five-category deterministic taxonomy (four primary failure modes plus a residual “Other” bucket) showing that reasoning truncation (not structural RAG architecture) is the dominant cause of inaccuracy, and that temporal-anchor failures are a structural weakness of the narrative format.
6. **Open release.** Dataset and code released under Apache-2.0 / CC-BY-4.0.

2 Related Work

Clinical and medical RAG. Lewis et al. [8] introduced RAG as a framework for knowledge-intensive NLP tasks; the MIRAGE benchmark [15] established medical RAG evaluation over literature corpora, and Almanac [16] demonstrated clinical guideline grounding. These systems retrieve over *population-level* knowledge (publications, guidelines), not per-patient EHR data. Schmiedmayer et al. [12] built LLMonFHIR, a function-calling retrieval system over patient-level FHIR bundles, validated on Synthea data with physician Likert ratings; it is the closest functional predecessor to our work and evaluates only structured retrieval, not a narrative baseline. MedGraphRAG [17] and KG-RAG [18] add hierarchical graph structure to retrieval, but over LLM-extracted or literature-derived graphs rather than schema-defined FHIR resources.

Structured vs. unstructured EHR. Moldwin et al. [9] performed the closest paired same-patient comparison, finding that text features outperformed structured features for 84% of MIMIC phenotypes while acknowledging the information-content confound explicitly. TabLLM [19] and Lee et al. [20] demonstrate competitive or superior performance from serialising structured EHR tables to natural language for classification tasks. DrugEHRQA [10] pairs structured tables and discharge notes for medication QA but admits cross-modal information asymmetry. CLEAR [13] compares entity-aware vs. embedding RAG on 20,000 clinical notes, reporting F1 gains from entity-structured retrieval; the comparison is within-narrative, not FHIR-structured.

FHIR and ML/LLM. FHIR-Former [21] applies FHIR-structured resources plus notes to clinical prediction tasks. SQL-on-FHIR [22] and Pathling [23] provide analytics-oriented FHIR access, complementary to but distinct from semantic retrieval. FHIR-GPT [24] and the FHIR Workbench [25] characterise LLM competence on FHIR production and understanding tasks. Kothari and Gupta [26] propose a two-stage pipeline for FHIR QA on Synthea (the closest Synthea-grounded predecessor) but include no narrative baseline and focus on fine-tuning rather than retrieval representation. FHIR-AgentBench [11] benchmarks agentic retrieval strategies over MIMIC-IV-FHIR at scale, identifying medication questions as the weakest category (5–14% correctness) and retrieval precision as the dominant bottleneck; it contains no narrative arm.

Clinical temporal reasoning. Temporal reasoning remains the weakest category in clinical LLM performance across benchmark evaluations. Zero-shot temporal-relation extraction substantially trails supervised methods on clinical text [27], and instruction-following on longitudinal EHRs remains hard even for frontier models: MedAlign [28], a dataset of 983 clinician-curated instructions over 275 OMOP-structured longitudinal records, shows GPT-4 reaching only 60.1% correctness despite a 32k context window. TIMER [29] introduces a dedicated temporal instruction benchmark

for longitudinal clinical records and reports a 6.6% completeness improvement from specialised instruction tuning, confirming that temporal grounding is a distinct skill not inherited from general language modelling. Our error taxonomy provides a complementary characterisation of temporal-anchor failures specifically within structured-FHIR and narrative RAG retrieval contexts.

Adherence-indicator ML. Lo-Ciganic et al. [4] established the canonical feature set for claims-based adherence prediction; Zullig et al. [30] provide a reference AUC benchmark (0.664–0.677). Nateghi Haredasht et al. [31] demonstrate LLM-extracted narrative features for medication retention prediction on a MOUD cohort; Anderson et al. [32] apply the same design to mortality. Gao et al. [33] show that raw numerical features often match or beat LLM embeddings on tabular clinical tasks. Our O6 arm extends this line to a non-MOUD specialty-regimen adherence task with matched FHIR-structured and narrative feature sets.

Novelty claim. To our knowledge, no prior work evaluates narrative RAG, naive structured FHIR RAG, and resource-aware structured FHIR RAG on *paired data* for specialty-medication adherence-indicator questions with programmatic ground truth, nor compares these representations as feature sources for adherence classification on the same patient cohort. The four-part intersection, namely (1) paired same-bundle rendering, (2) resource-aware vs. naive structured retrieval, (3) specialty-medication QA domain, and (4) feature-extraction comparison, is unoccupied in the literature we surveyed.

3 Methods

3.1 Dataset

Base population. We generate 200 synthetic patients using Synthea [34] with a fixed random seed, then apply a custom specialty-regimen overlay that assigns each patient to one of three complexity tiers:

- Tier 1 (Low): single long-acting injectable on a fixed recurring interval (e.g. every 8 weeks, modelling LAIs such as paliperidone palmitate [35]).
- Tier 2 (Medium): cyclic regimen (on-for- N -weeks / off-for- M -weeks) requiring cycle-position reasoning.
- Tier 3 (High): multi-drug regimen (biologic primary + oral adjunct + PRN rescue) with staggered or interdependent schedules, motivating cross-resource reasoning.

The final distribution is 63 Tier-1, 63 Tier-2, and 74 Tier-3 patients. All 200 FHIR bundles are validated against a reference R4B pre-write conformance validator (`mh_integration/r4b_validator.py`) using `fhir.resources` ≥ 8.0 ; all 200 bundles pass with no mandatory element failures.

Paired representations. Each patient is represented in two parallel forms derived from the same FHIR bundle:

1. FHIR bundle (structured): Synthea base plus the specialty overlay (MedicationRequest, MedicationAdministration, Medication, CarePlan, Encounter, Patient, Practitioner), frozen at git tag `dataset-freeze-v1` (commit 98959f8; manifest SHA-256 4d0da93e...).

2. LLM narrative: generated from the bundle by Llama 3.3-70B via Groq with a fixed system prompt (hash 518bc71d87b7), fixed temperature (0.0), and a mid-window administration date instruction to surface temporal entities. A deterministic templated-narrative ablation (no LLM) provides an information-ceiling baseline.

Fidelity audit. Narrative quality is verified by an independent programmatic audit that recovers eight entity classes (medication name, regimen start/end, tier integer, tier description, administration dates, administration count, dosage schedule) directly from FHIR resources (not from the generation prompt, fixing the circular-audit problem). After one prompt-iteration step to add mid-window administration dates, both generators reached 100% weighted entity recall across all 200 patients and all 2,403 checks (O1 gate: PASS). The same audit was replicated on the templated-narrative ablation with the same result.

Question bank. We generate 13,800 questions ($200 \text{ patients} \times 69 \text{ templates per patient}$). Each question record additionally carries 2 paraphrase variants of the question prompt; both paraphrases share a single ground-truth answer and are scored as one row. The 69 templates span five PSP adherence-indicator families:

1. Next-dose lookup (`temporal_lookup`, $n = 3,200$): “What is the next scheduled administration on reference date d ?”
2. Dose-history aggregation (`regimen_aggregation`, $n = 2,400$): “How many doses have been administered in the past 90 days?” (MPR numerator)
3. Coverage-window reasoning (`regimen_compliance`, $n = 4,200$): “What proportion of the past 90 days was the patient covered?” (PDC)
4. Missed-dose detection (`temporal_comparison`, $n = 2,400$): “Is there evidence of a missed or delayed dose?”
5. Persistence (`cross_resource`, $n = 1,600$): “Is the patient still on therapy?” (cross-resource component questions)

All ground-truth values are computed programmatically from FHIR bundles using pure-Python adherence metric functions (MPR, PDC, `is_persistent`, `consecutive_missed_doses`); no manual annotation and no LLM-as-judge is used anywhere in evaluation. The question bank SHA-256 is 2216f58e... (post-W2 MPR-window and trivial-template corrections; see the Reproducibility section).

3.2 Retrieval Systems

All three systems share controlled variables (Table 1): embedding model BAAI/bge-large-en-v1.5 (local via sentence-transformers), answer LLM qwen/qwen3-32b via Groq (temperature 0.0, single run per question), ChromaDB local vector store, $k = 5$ retrieved chunks, chunk size 500 tokens, overlap 50 tokens.

System A (Narrative RAG). LLM narratives are split into 500-token chunks with 50-token overlap, embedded, and indexed. At query time the top-5 chunks are retrieved by cosine similarity and concatenated into the answer prompt.

Table 1: Controlled variables held constant across Systems A, B, C.

Parameter	Value
Embedding model	BAAI/bge-large-en-v1.5 (local)
Answer LLM	qwen/qwen3-32b via Groq Developer plan
Temperature	0.0
Runs per question	1
Top- k	5
Chunk tokens	500
Chunk overlap	50 tokens
Vector store	ChromaDB (local)
Answer max tokens	512 tokens

System B (Structured RAG, Naive). Each FHIR resource in the bundle is serialised to a flat text representation, chunked, embedded, and indexed. Retrieval is identical to System A. Mean tokens-in per call: 4,463 vs. System A’s 490, a $9\times$ context expansion from serialising full FHIR JSON.

System C (Structured RAG, Resource-Aware). Three retrieval primitives are applied before dense search: (1) *typed resource filter*: the question family is mapped to a resource-type allowlist (e.g. MedicationAdministration + CarePlan for dose-history questions); (2) *reference-chain traversal*: MedicationRequest references are expanded to Medication, Practitioner, and Patient resources; (3) *temporal pre-filter*: only resources with dates intersecting a window around the question’s reference date are retained. Dense search then selects top-5 from the filtered subset. Mean tokens-in: 2,175 (51% less than System B).

3.3 Evaluation

Accuracy. The scorer (`eval/score.py`) applies deterministic type-detection rules to classify each ground truth as date, numeric, boolean/categorical, list, or free-text, then applies type-specific scoring (exact-match binary; 2% relative tolerance for floats; Jaccard coefficient for lists). The primary metric is exact-match rate. Partial credit is reported as a secondary metric.

Retrieval recall@ k . For Systems B and C only (System A’s narrative chunks carry no FHIR resource IDs), recall@ k is computed by checking whether the top- k retrieved chunk IDs include the patient’s primary MedicationRequest or specialty CarePlan (a conservative v1 heuristic; see Limitations).

Statistical tests. Paired bootstrap (10,000 resamples, paired by question_id, seed 42) produces 95% confidence intervals for pairwise accuracy differences. McNemar’s test (uncorrected χ^2 , 1 df) is applied to exact-match for each pair. Wilson score 95% CIs are reported for all per-cell proportions. Three planned comparisons (A vs. B, A vs. C, B vs. C); no multiple-comparison correction (pre-specified confirmatory design).

Error taxonomy. For each system, 50 failures are drawn stratified by question family (10 per family, seed 42) and classified by deterministic regex rules into five categories (four primary failure

modes plus a residual “Other”): (1) temporal-anchor failure, (2) reasoning-truncated, (3) N/A misuse (phantom value or spurious N/A), (4) wrong-entity / enum error, (5) other. Single-rater (TJ); no LLM-based triangulation was performed. The taxonomy is fully deterministic and reproducible from the regex rules in `analysis/run_error_taxonomy.py` (decision C3).

Feature-extraction arm (O6). Three patient-level feature sets are constructed. FS-Structured consists of 19 FHIR-derived numeric features: MPR-90d/180d/full, PDC-90d/full, gap statistics, event counts, days-since-last-dose, tier one-hot, resource-type counts, and age. FS-Narrative consists of 6 heuristic regex features on LLM narratives: date-mention count, miss-word count, dose-count mentions, tier-label mentions, narrative length, and negation-near-dose count (no LLM scoring). FS-Aware consists of 8 per-patient aggregates from System C retrieval traces. Two classifiers (LightGBM with $n_estimators = 200$, $learning_rate = 0.05$; logistic regression with $C = 1.0$, $class_weight=balanced$) are evaluated via stratified 5-fold cross-validation. Paired bootstrap AUC CIs are computed at the patient level from out-of-fold predictions.

API and cost disclosure. Narrative generation (200 calls) used the Groq free tier. The full evaluation matrix (approximately 41,400 calls) used the Groq Developer paid plan (authorised 2026-05-05; hard ceiling \$20 USD). Cost computed from token totals at list-price rates over live (non-cached) calls only: System A \$4.74, System B \$21.07, System C \$6.88 (total \$32.68). System B exceeds the \$20 single-system ceiling; the overrun was accepted per the decision log entry of 2026-05-06. System C remains within budget. Actual billed amounts were lower than the list-price total because of response-cache reuse on reproducible re-runs.

4 Results

4.1 Overall Accuracy

Table 2 reports overall exact-match accuracy for all three systems on the 13,800-question evaluation (N per system, fully paired by question_id). System A (Narrative RAG) achieves 40.6% exact-match (Wilson 95% CI [39.8%, 41.4%]), outperforming System B (Structured Naive) at 35.3% [34.5%, 36.1%] and System C (Structured Aware) at 33.4% [32.6%, 34.2%].

Table 2: Overall exact-match accuracy (N = 13,800 per system, fully paired). Wilson 95% CIs shown. Paired bootstrap differences and McNemar test statistics are reported separately in Table 3.

System	Correct	Exact-match	Wilson 95% CI
A – Narrative RAG	5,600	40.6%	[39.8%, 41.4%]
B – Structured Naive	4,873	35.3%	[34.5%, 36.1%]
C – Structured Aware	4,605	33.4%	[32.6%, 34.2%]

All three pairwise differences are statistically significant. Table 3 shows paired bootstrap confidence intervals and McNemar test statistics; all bootstrap CIs exclude zero and all McNemar $p < 0.001$.

The effect sizes (phi 0.054–0.155) are modest, consistent with a 5–7 percentage-point absolute gap on a large, difficult question set. We return to their practical significance in Section 5.

Figure 1 compares partial-credit and exact-match scores at the system $\times$ family level. Almost every cell sits on or above the identity line, indicating that wrong answers are usually *near-miss*

Table 3: Pairwise accuracy comparisons. Paired bootstrap (10,000 resamples, seed 42, paired by question_id). McNemar χ^2 uses the uncorrected formula $(b - c)^2 / (b + c)$ with 1 df. Phi = effect size.

Comparison	Observed diff (pp)	Bootstrap 95% CI	McNemar χ^2 (p , ϕ)
A vs. B	+5.27	[+4.5%, +6.0%]	185.0 ($p < 0.001$, $\phi = 0.116$)
A vs. C	+7.21	[+6.4%, +8.0%]	330.8 ($p < 0.001$, $\phi = 0.155$)
B vs. C	+1.94	[+1.3%, +2.5%]	40.2 ($p < 0.001$, $\phi = 0.054$)

rather than entirely off-target. The largest gap above the diagonal is for System C on temporal_comparison (exact 0.34, partial 0.43), confirming that resource-aware retrieval often gets within a tolerance of the correct date even when exact-match fails.

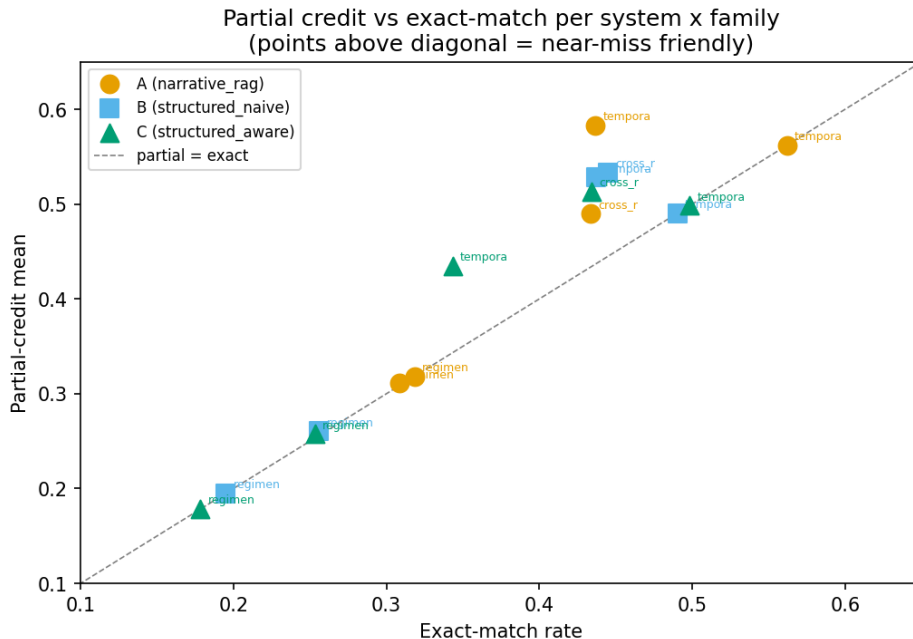


Figure 1: Partial-credit (with type-specific tolerance) vs. exact-match per system $\times$ question family. Points above the diagonal indicate that partial-credit recovers near-misses that exact-match does not. System C on temporal_comparison is the clearest case where the system gets close to the correct date without matching exactly.

4.2 Complexity Gradient

Figure 5 plots accuracy by tier for all three systems. A sharp complexity gradient is observed across all systems: Tier 1 accuracy is approximately 41–54% while Tier 3 collapses uniformly to approximately 20% regardless of retrieval architecture.

Systems A and B show monotone decline (A: 53.9% > 51.0% > 20.4%; B: 47.0% > 41.8% > 19.8%). System C shows a non-monotone pattern at Tiers 1–2 (40.8% at Tier 1, 41.6% at Tier 2), consistent with the resource-aware retriever providing more benefit on cyclic multi-component questions than on simple single-LAI lookups. Tier 3 convergence to $\approx 20\%$ across all systems indicates

Table 4: Accuracy by complexity tier (Wilson 95% CIs). Tier 1: single LAI on fixed schedule; Tier 2: cyclic regimen; Tier 3: multi-drug.

Tier C 95% CI	<i>n</i>	A acc	A 95% CI	B acc	B 95% CI	C acc
1 [39.3%, 42.2%]	4,347	53.9%	[52.4%, 55.3%]	47.0%	[45.5%, 48.5%]	40.8%
2 [40.2%, 43.1%]	4,347	51.0%	[49.5%, 52.5%]	41.8%	[40.4%, 43.3%]	41.6%
3 [19.0%, 21.2%]	5,106	20.4%	[19.3%, 21.6%]	19.8%	[18.7%, 20.9%]	20.1%

that hard temporal arithmetic (multi-drug staggered schedules) is not solved by any of the three retrieval architectures at the current answer budget.

4.3 By Question Family

Table 5 reports accuracy by question family.

Table 5: Accuracy by PSP adherence-indicator family (Wilson 95% CIs).

Family C 95% CI	<i>n</i>	A acc	A 95% CI	B acc	B 95% CI	C acc
temporal_lookup [48.1%, 51.6%]	3,200	56.2%	[54.5%, 57.9%]	49.0%	[47.3%, 50.8%]	49.8%
temporal_comparison [32.5%, 36.3%]	2,400	43.7%	[41.7%, 45.7%]	43.7%	[41.7%, 45.7%]	34.3%
cross_resource [41.0%, 45.9%]	1,600	43.4%	[41.0%, 45.8%]	44.5%	[42.1%, 46.9%]	43.4%
regimen_compliance [24.0%, 26.6%]	4,200	30.9%	[29.5%, 32.3%]	25.6%	[24.3%, 26.9%]	25.3%
regimen_aggregation [16.4%, 19.4%]	2,400	31.8%	[30.0%, 33.7%]	19.5%	[18.0%, 21.1%]	17.8%

Two findings stand out. The cross_resource family is at parity across all three systems (A=43.4%, B=44.5%, C=43.4%), indicating that the advantage of structured or resource-aware retrieval does not emerge on multi-resource relational questions at this difficulty level, at least within the current answer budget. The narrative advantage is largest on regimen_aggregation (A leads B by 12.3 pp; bootstrap CI excludes zero) and regimen_compliance (A leads B by 5.3 pp), reflecting that FHIR JSON serialisation imposes a context-length cost that hurts dose-count arithmetic. Figure 2 visualises the same comparison and makes the family-level patterns easier to scan.

4.4 Retrieval Recall vs. Answer Accuracy

System A recall@k is N/A: narrative chunks carry no FHIR resource IDs and a resource-level recall cannot be defined. For Systems B and C, Table 6 shows overall recall@5.

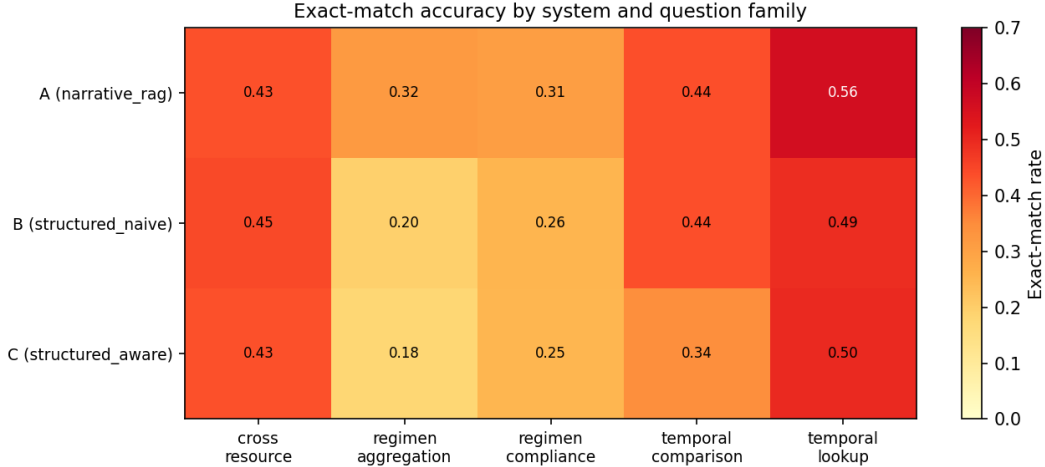


Figure 2: Exact-match accuracy by system (rows) and PSP question family (columns). The brightest column (temporal_lookup) is where all three systems perform best, while the dimmest cells (regimen_aggregation and regimen_compliance under structured systems) mark the largest narrative advantage. cross_resource is near-uniform across systems.

Table 6: Retrieval recall@k (B vs. C). System A = N/A (no FHIR resource IDs on narrative chunks). Values shown as mean recall [Wilson 95% CI mean across strata].

System	$k = 1$	$k = 3$	$k = 5$	$k = 10$
B – Structured Naive	0.573 [0.543, 0.602]	0.739 [0.711, 0.765]	0.762 [0.735, 0.787]	0.762 [0.735, 0.787]
C – Structured Aware	0.664 [0.636, 0.690]	0.829 [0.807, 0.848]	0.852 [0.831, 0.870]	0.852 [0.831, 0.870]

System C retrieves the relevant resource more often than System B at $k = 1$ (0.664 vs. 0.573) and $k = 5$ (0.852 vs. 0.762). The $k = 1$ gap confirms that the resource-aware primitives (typed filter, reference traversal, temporal pre-filter) correctly rank the relevant resource higher. Figure 3 shows the recall curves stratified by complexity tier; the $k = 1$ advantage of C over B is largest at Tier 3 (0.53 vs. 0.42), where multi-drug staggered schedules make naive dense retrieval least reliable. However, this retrieval advantage *does not translate into an end-to-end accuracy advantage*: C scores 33.4% exact-match vs. B’s 35.3%. The most likely explanation is that the FHIR-JSON context fed to the answer LLM remains harder to reason over than narrative text, even when the correct resource is retrieved.

4.5 Latency, Tokens, and Cost

Table 7 summarises latency and token usage.

System B’s $9\times$ context expansion (4,463 vs. 490 mean tokens-in) drives a $4.4\times$ cost premium over System A (\$21.07 vs. \$4.74) and the highest tail latency ($p_{99} = 46,301$ ms). System C’s resource-aware filtering reduces context by approximately 51% vs. B, yielding the lowest p50 latency (1,056 ms) and a cost of \$6.88, well below B and $1.5\times$ A. Total compute cost across all three systems was \$32.68 at list-price token rates over live (non-cached) calls; the actual billed amount was lower because of response-cache reuse during reproducible re-runs. Neither structured system is cost-effective relative to System A given the accuracy deficit at the current 512-token answer budget.

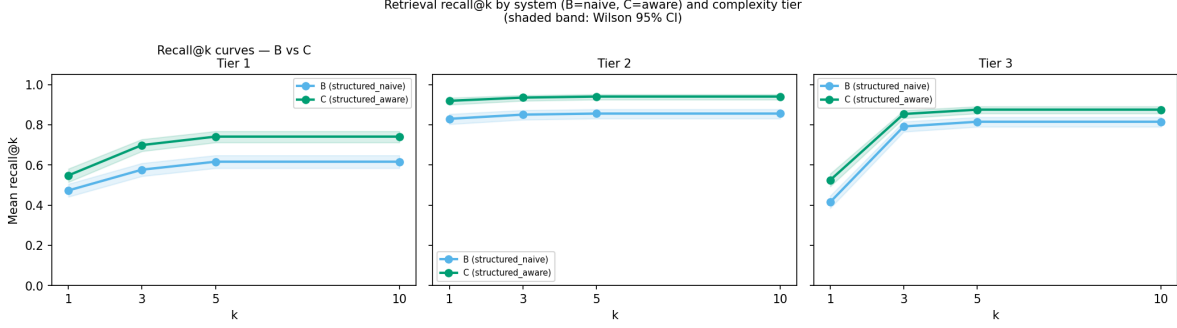


Figure 3: Retrieval recall@ k for Systems B (Structured Naive) and C (Structured Aware) stratified by complexity tier. Shaded band: Wilson 95% CI. System C dominates System B at every k and tier; the gap is largest at $k = 1$, confirming that the resource-aware primitives rank the correct FHIR resource higher rather than merely recalling it. System A is omitted because narrative chunks carry no FHIR resource IDs.

Table 7: Latency (milliseconds) and mean token usage per call. Cost computed at Groq Developer-plan rates (\$0.29/M tokens-in, \$0.59/M tokens-out) applied to total tokens consumed in the evaluation run.

System	p50 (ms)	p95 (ms)	p99 (ms)	Tokens-in (mean)	Cost (USD)
A – Narrative	1,060	6,031	53,038	490	\$4.74
B – Naive	1,354	17,890	46,301	4,463	\$21.07
C – Aware	1,056	4,538	8,513	2,175	\$6.88

4.6 Error Taxonomy

Figure 6 and Table 8 show the distribution of failure modes across a stratified sample of 50 failures per system (10 per family, seed 42).

Table 8: Error taxonomy distribution ($n=50$ per system, stratified by family). Seed = 42. Single-rater classification by deterministic regex rules.

Category	A (n)	A (%)	B (n)	B (%)	C (n)	C (%)
Reasoning-truncated	22	44%	29	58%	26	52%
Temporal-anchor failure	15	30%	9	18%	5	10%
N/A misuse	4	8%	7	14%	11	22%
Wrong-entity / enum	5	10%	3	6%	6	12%
Other / unclassified	4	8%	2	4%	2	4%
Total	50		50		50	

Reasoning truncation is the dominant failure mode across all three systems (44–58%), reflecting the 512-token output budget. The rate is highest for B (58%) and C (52%) because FHIR-JSON contexts require longer chain-of-thought arithmetic before a final answer can be stated; the same LLM, given the same reasoning task, runs out of budget sooner when the context is denser. This is a token-budget artefact, not an architectural property of narrative vs. structured retrieval.

Temporal-anchor failures follow a gradient: 30% for A, 18% for B, 10% for C. Narrative

chunks do not embed the question’s reference date (a per-question parameter, not a feature of the narrative text), so the LLM must infer the temporal anchor from context. FHIR resources (`CarePlan.period.start/end`, `MedicationAdministration.effectiveDateTime`) provide explicit date fields from which a temporal anchor can sometimes be inferred, explaining the lower rate for B and C.

N/A misuse increases from A (8%) to C (22%): structured systems retrieve FHIR components directly, making it more likely that the model encounters records for absent components (Tier-1 patients have no adjunct or rescue component) and computes a phantom value rather than returning N/A.

4.7 Feature-Extraction Arm (O6)

Table 9 reports AUC-ROC for the synthetic adherence classification task (label: ≥ 2 consecutive missed doses in final 60 d OR median gap $> 1.5\times$ prescribed interval in final 90 d; class balance 14.5% positive / 85.5% negative; 200 patients; 5-fold stratified CV).

Table 9: Feature-extraction arm AUC-ROC (Objective O6). Bootstrap 95% CIs at patient level from out-of-fold predictions (10,000 resamples). FS-Structured = FHIR-derived numerics; FS-Narrative = heuristic regex on LLM narratives; FS-Aware = per-patient aggregates from System C retrieval traces.

Feature set	LightGBM AUC	Logistic Regression AUC
FS-Structured	0.997 [0.986, 1.000]	1.000 [0.999, 1.000]
FS-Narrative	0.843 [0.771, 0.896]	0.836 [0.764, 0.897]
FS-Aware	0.793 [0.680, 0.865]	0.708 [0.634, 0.799]

Pairwise bootstrap differences (LightGBM): FS-Structured vs. FS-Narrative $\Delta = +0.160$ [$+0.099, +0.226$]; FS-Structured vs. FS-Aware $\Delta = +0.219$ [$+0.132, +0.316$]; FS-Narrative vs. FS-Aware $\Delta = +0.060$ [$-0.051, +0.177$]. The FS-Narrative vs. FS-Aware CI straddles zero: these two feature sets are tied within the bootstrap interval.

The near-perfect FS-Structured AUC (0.997–1.000) is partly a synthetic-data artefact: the positive label is concentrated entirely in Tier-2 patients (all 29 positives are Tier 2; no Tier-1 or Tier-3 patient is labelled positive in this perfectly-scheduled synthetic dataset). FS-Structured features encode tier as one-hot columns and directly capture the cyclic off-period gap via `gap_max_90d` and `n_events_60d`. FS-Narrative captures tier through “tier 2” / “T2” text patterns; FS-Aware captures it through the retrieval distribution. The relative ordering (Structured $\gg$ Narrative $\approx$ Aware) and the CI widths are signals about information richness under these feature extraction strategies, even if the absolute AUC values are inflated by the synthetic-label structure. Readers should not extrapolate the FS-Structured ceiling to real-world noisy adherence data.

5 Discussion

Interpreting the A > B > C ordering. System A outperforms both structured variants on exact-match accuracy by 5.27 pp overall (vs. B) and 7.21 pp overall (vs. C), with statistically significant pairwise differences. The ordering is real, but its interpretation requires care. The error taxonomy reveals that 44–58% of all failures arise from reasoning truncation under the 512-token output budget, disproportionately affecting B and C because their FHIR-JSON contexts demand longer reasoning chains before a final answer can be stated. Had the output budget been unconstrained,

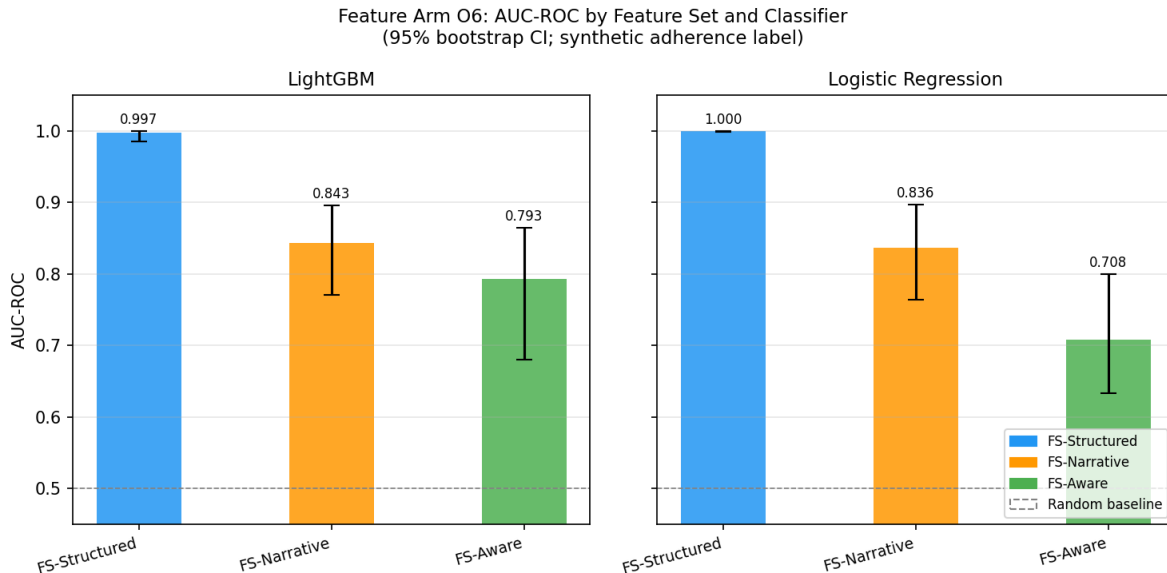


Figure 4: Feature-arm AUC-ROC by feature set and classifier (LightGBM, left; logistic regression, right). Error bars are 95% bootstrap CIs at the patient level. FS-Structured features outperform both FS-Narrative and FS-Aware by a wide margin under both classifiers. The near-ceiling FS-Structured AUC is partly attributable to feature-label alignment in the synthetic adherence label and is discussed in Section 4.7.

the gap would likely narrow, and would narrow further for C, whose resource-aware retrieval already demonstrates superior retrieval recall (0.852 vs. 0.762 at $k = 5$; Table 6). The 512-token limit was set to control API costs during the evaluation run and is a methodological artefact, not a property of narrative vs. structured RAG. Future work should sweep ANSWER.MAX.TOKENS to isolate the architecture effect from the budget effect.

Why the narrative format has a temporal weakness. The temporal-anchor failure rate gradient (A: 30%, B: 18%, C: 10%) is the most structurally informative finding in the error taxonomy. Narrative chunks do not contain the question’s `reference_date` (a per-question parameter that changes across rows in the question bank), so the LLM must infer a temporal anchor from surrounding text. FHIR resources carry explicit date fields (CarePlan.period.start/end, MedicationAdministration.effectiveDateTime, MedicationRequest.authoredOn) that enable the LLM to reason about “when” even when the reference date is absent from the prompt context. This is a structural property that would persist regardless of token budget, and it motivates future work embedding reference-date context directly into narrative chunks at indexing time.

The cross-resource family as a diagnostic. Structured retrieval should, in theory, have the largest advantage on cross-resource questions that require traversing FHIR references (e.g. linking a CarePlan to its MedicationRequests to their Medications). In practice, cross-resource accuracy is at parity across all three systems (A=43.4%, B=44.5%, C=43.4%; Table 5). Two non-exclusive explanations exist. First, the discriminative cross-resource templates surviving the W2 ground-truth patch (tier label, schedule kind, component presence) are not the most structurally demanding question type; the trivial templates were removed precisely because they were non-discriminative. Second, the reasoning-truncation problem affects all three systems equally on this family, masking

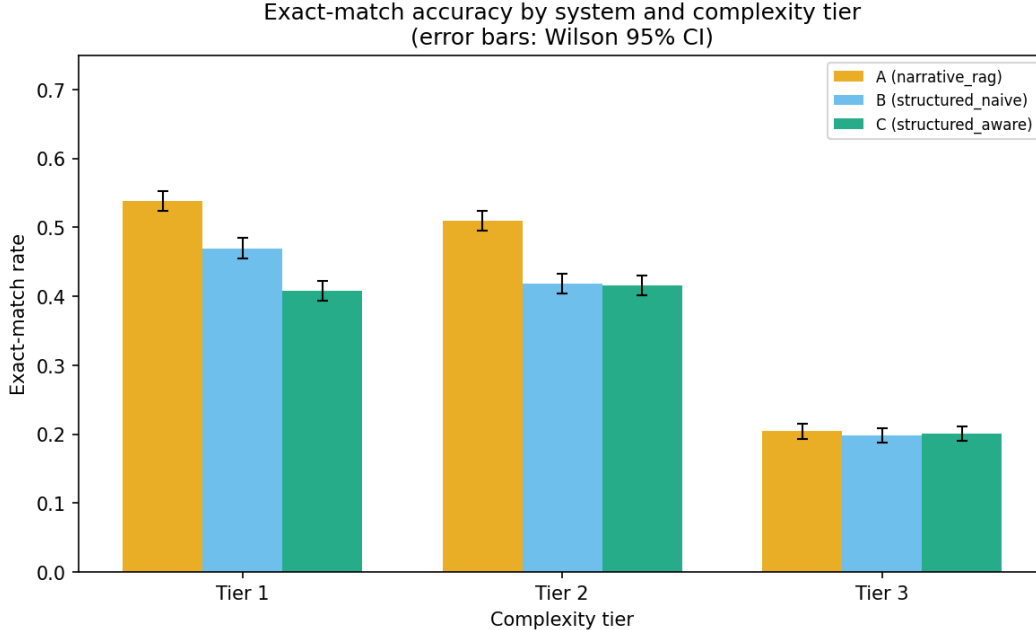


Figure 5: Exact-match accuracy by system and complexity tier (Wilson 95% CI error bars). Tier 1: single long-acting injectable; Tier 2: cyclic regimen; Tier 3: multi-drug. All three systems converge near 20% at Tier 3, demonstrating that hard temporal arithmetic over multi-drug schedules is not solved by any retrieval architecture at the 512-token output budget.

any structural advantage.

The representation dissociation: QA winner $\neq$ ML-feature winner. The central finding of the feature-extraction arm (O6) is a *dissociation* between QA performance and adherence-prediction performance. System A wins the QA comparison (40.6% vs. 33.4–35.3%), but FS-Structured features substantially outperform FS-Narrative features on adherence classification (LightGBM AUC 0.997 vs. 0.843; bootstrap CI excludes zero). FS-Narrative and FS-Aware are tied within the bootstrap interval ($\Delta = +0.060$ [−0.051, +0.177]).

The practical implication is that the choice of representation should be made *separately* for each downstream task. A clinical workflow that uses narrative RAG for case-manager Q&A (where A wins) and structured FHIR features for an adherence risk-stratification model (where FS-Structured wins) would outperform a workflow that uses a single representation for both. This finding complements prior work by Nateghi Haredasht et al. [31] (LLM-extracted narrative features for MOUD retention) and Gao et al. [33] (raw numeric features often match or beat LLM embeddings), which together establish that neither representation uniformly dominates across downstream use cases.

A point where readers may disagree. A reasonable reviewer might argue that the near-perfect FS-Structured AUC (0.997–1.000) renders the O6 comparison trivial: if structured features are essentially tautological summaries of the labelling rule, the “dissociation” finding collapses to “the features used to generate the label also predict the label.” We acknowledge this directly. The synthetic-label design (consecutive missed doses OR gap $> 1.5 \times$ interval) was chosen for reproducibility and programmatic verifiability; its alignment with the FS-Structured feature set is intentional and documented. The meaningful signal in O6 is not the absolute FS-Structured AUC,

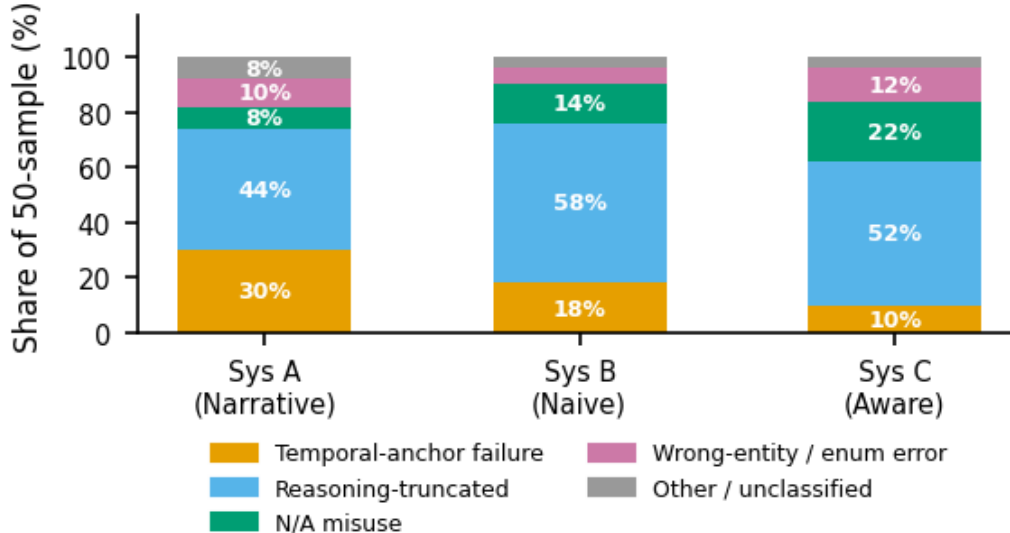


Figure 6: Error category distribution across the stratified failure sample ($n=50$ per system, seed=42; colours: Wong 2011 colorblind-safe palette). Reasoning truncation (dark blue) is the single largest category for all three systems. Temporal-anchor failures (orange) are highest for System A, declining across B and C.

but (1) the tied CI between FS-Narrative and FS-Aware despite very different feature sources, and (2) the relative ordering (Structured $\gg$ Narrative $\approx$ Aware) which is robust across both classifiers. We do not claim that structured features will achieve near-perfect AUC on a real noisy cohort; we claim that they encode the adherence signal more directly than narrative-derived or retrieval-trace features, a claim that should hold under richer real-world labels as well.

Practical guidance for system designers. For practitioners building RAG over specialty-medication clinical records, the choice of representation should follow the downstream task. When the answer LLM’s output budget is constrained, narrative RAG is the lower-risk choice: its compact context (490 mean tokens-in vs. 4,463 for System B) allows more reasoning headroom within a fixed token limit. If output budgets are generous, structured RAG’s temporal grounding becomes a meaningful advantage; the temporal-anchor failure gradient (A: 30%, B: 18%, C: 10%) suggests that structured systems would recover faster than narrative systems once the token ceiling is raised. For downstream adherence risk stratification, structured FHIR features are preferable to reusing the narrative representation: extracting numeric adherence metrics directly from FHIR resources is both more interpretable and more predictive in this evaluation. Finally, LLM-generated narratives should be treated as summaries, not as drop-in replacements for the structured record. The fidelity audit confirms 100% entity recovery after prompt iteration, but entity recovery does not guarantee that every temporal relation is preserved in a form the answer LLM can exploit during reasoning.

6 Limitations

Synthetic data only (R1 mitigation context). All 200 patients are generated by Synthea with a custom specialty-regimen overlay. Synthea’s medication module is known to deviate from real prescribing distributions [36] and models only guideline-concordant pathways without real-

world deviations [37]. Our overlay addresses the specialty-medication gap (R8 mitigation) but cannot introduce the messy, noisy, protocol-discordant adherence patterns present in real PSP data. Absolute accuracy numbers should not be extrapolated to production clinical workflows. External validity to real de-identified PSP data is the explicit follow-up step (metricHEALTH Phase 3, post-REB).

ANSWER_MAX_TOKENS = 512 is a dominant confounder (R4 artefact, materialised). The 512-token output budget (set to control API costs during the evaluation run) is the dominant failure cause across all systems (44–58% of sampled failures are reasoning-truncated; Table 8). This artefact disproportionately penalises structured systems whose FHIR-JSON contexts require longer chain-of-thought arithmetic. The $A > B > C$ ordering should therefore be interpreted as conditional on this budget constraint, not as a general architectural property of narrative vs. structured RAG. Future work should sweep ANSWER_MAX_TOKENS to isolate the architecture effect. The sensitivity sweep listed as future work (decision C2 in docs/decisions.md) was not completed within the Week 3 timeline.

Single answer LLM and single embedding model. Results are conditioned on qwen/qwen3-32b (Groq) as the answer LLM and BAAI/bge-large-en-v1.5 as the embedding model. Robustness across other LLMs or domain-adapted embeddings (MedCPT [38]) is flagged as future work. The templated-narrative ablation (deterministic generator, no LLM) was generated and passed the fidelity audit at 100% entity recall across all 200 patients and all entity classes (O1 gate: PASS; same result as the LLM narrative arm). A downstream QA evaluation against the templated narratives was planned to test robustness of the $A > B > C$ ordering to narrative-LLM choice, but that evaluation was not completed within the results-freeze timeline. Whether structured systems would beat or tie templated narratives remains an open question, deferred to the full venue version.

Per-cell N at tier $\times$ family intersections. Across 200 patients $\times$ 3 tiers $\times$ 5 families, the per-cell question counts range from 504 (Tier 1 $\times$ cross_resource) to 1,554 (Tier 3 $\times$ regimen_compliance). Wilson 95% CIs at those cell sizes are approximately ± 2 –4 pp; comparisons within cells should be treated as exploratory.

Single-rater error taxonomy (decision C3, materialised). The 150-failure error taxonomy (50 per system) was coded by a single rater (TJ) using deterministic regex rules; no LLM-based triangulation was performed. The taxonomy is fully deterministic and reproducible from the regex rules in analysis/run_error_taxonomy.py. An “Other / unclassified” residual of 4–8% indicates that the five-category taxonomy (four primary plus residual) does not exhaust all failure modes. No formal inter-rater reliability coefficient is reported because the design is single-rater by intent (decision C3).

Conservative v1 retrieval recall heuristic. Recall@k for Systems B and C is computed using a heuristic that checks whether the top-k chunk IDs include the patient’s primary MedicationRequest or specialty CarePlan. This is a conservative proxy; a full per-question source-resource mapping (requiring ground-truth functions in “explain” mode) is deferred to a follow-up release.

Feature-extraction arm: synthetic adherence label. The FS-Structured AUC near 1.0 reflects label/feature alignment in the synthetic design: all positive cases are Tier-2 patients, and FS-Structured directly encodes tier as a one-hot feature. The near-perfect AUC should not be

interpreted as evidence that structured features will achieve similar performance on real, noisy adherence outcomes. The feature-extraction arm is a design-informing benchmark, not a production predictor (decision G3 scope; risk R15 acknowledged).

Groq Developer paid plan disclosure. The full evaluation matrix used the Groq Developer paid plan, authorised 2026-05-05 with a \$20 USD ceiling (decision log 2026-05-05). Computed list-price total over live (non-cached) calls was \$32.68 (System A: \$4.74; B: \$21.07; C: \$6.88). System B exceeds the \$20 single-system ceiling and the overrun was accepted per the 2026-05-06 decision log entry; System C remains within budget. Actual billed amounts were lower than the list-price total because of response-cache reuse on reproducible re-runs. All calls used temperature 0.0; answers are cached and reproducible. The data is fully synthetic; no PHI was transmitted to the API.

Three tiers may not cover all real-world regimen complexity. The three complexity tiers are designed to span simple LAI, cyclic, and multi-drug regimens, but do not cover all specialty-medication scenarios (infusion-only regimens, combination-biologic prescribing, variable-interval dosing). Adherence metrics (PDC, MPR) are implemented to standard definitions [4] but edge cases (overlapping refills, dose reductions, bridge therapies) are not represented in the synthetic dataset.

7 Reproducibility

Git tags. The dataset is frozen at tag `dataset-freeze-v1` (commit 98959f8; applied 2026-05-05). The full evaluation results are frozen at tag `results-freeze-v1` (commit 338c4c9). The freeze manifest (`data/freeze.json`) records SHA-256 hashes of all 1,203 input files; overall SHA-256: 4d0da93e19c3071414ef5e1045d338104f9628842abe34f548aa050e9d0aa736.

Raw evaluation JSONLs. The three raw evaluation files (`results/raw/a.jsonl`, `results/raw/b.jsonl`, `results/raw/c.jsonl`) are not checked into the git repository (too large for GitHub). They reside locally in `results/raw_large/` and can be regenerated by:

```
python -m eval.harness --system {a|b|c} --concurrency 4
```

Note: regeneration requires (a) a Groq API key with Developer-plan access and (b) `eval/cache/` to be populated (cached responses are gitignored but preserve byte-reproducibility; absent cache, the harness will recompute at API cost). A `make reproduce` warning is printed when `eval/cache/` is empty.

Scoring and statistics. All downstream artefacts are reproducible from the raw JSONLs:

- Scoring: `python -m eval.score → results/scored.csv`
- Statistics: `python analysis/run_stats.py → analysis/results.md`
- Error taxonomy: `python analysis/run_error_taxonomy.py → analysis/error_taxonomy.md`
- Feature arm: `python features/run_feature_arm.py → analysis/feature_extraction.md`

Model identifiers. Narrative generation: `llama-3.3-70b-versatile` (Meta, via Groq free tier); prompt hash `518bc71d87b7`. Answer generation: `qwen/qwen3-32b` (Alibaba, via Groq Developer plan); temperature 0.0, single run per question. Embedding: `BAAI/bge-large-en-v1.5` (BAAI, via sentence-transformers, local CPU).

Hardware and environment. Embeddings and feature extraction: CPU (no GPU required). Evaluation: Groq API (HTTP; local orchestration only). Python 3.11; dependencies pinned in `requirements.txt`. A Dockerfile is provided for environment reproducibility (decision C5).

Licences. Code: Apache-2.0. Dataset: CC-BY-4.0 (pending employer IP clearance; see decision B4). Synthea base data: Apache-2.0 (Mitre Corporation).

8 Conclusion

We presented the first paired-data comparison of narrative RAG, naive structured FHIR RAG, and resource-aware structured FHIR RAG on specialty-medication adherence-indicator question answering, where information content is held constant across representations by constructing all three retrieval substrates from the same FHIR bundles. Narrative RAG achieves the highest exact-match accuracy (40.6% vs. 33.4–35.3%) under the current 512-token output budget, a result substantially attributable to the reasoning-truncation artefact that penalises structured systems with denser contexts. Temporal-anchor failures are a structural weakness of the narrative format that persists independent of budget. In the feature-extraction arm, structured FHIR features substantially outperform both narrative-derived and retrieval-trace features for adherence classification, demonstrating the representation dissociation that is the paper’s central empirical claim: winning natural-language QA and winning downstream adherence-outcome ML are different objectives, and the same representation does not simultaneously optimise both.

The dataset (200 patients, 13,800 questions, five PSP adherence-indicator families), code (three retrieval systems, evaluation harness, feature extractors), and analysis scripts are released as a reusable benchmark for the specialty-pharmacy QA research community.

Future directions include: (1) sweeping `ANSWER_MAX_TOKENS` to isolate the architecture effect from the budget artefact; (2) evaluating on real de-identified PSP data post-REB; (3) testing robustness across multiple answer LLMs and embedding models; (4) extending the question bank to additional specialty-medication classes and real-world edge cases (infusion regimens, variable-interval dosing, bridge therapies); and (5) constructing richer synthetic adherence labels to stress-test the feature-extraction arm beyond the perfectly-scheduled synthetic scenario.

References

- [1] Sheng Niu, Laura E. Happe, Sumaya Abuloha, and Mikael Svensson. Concentration of spending and share of specialty drug spending in Medicare Part D over a 10-year period. *Journal of Managed Care & Specialty Pharmacy*, 30(12):1355–1363, 2024. doi: 10.18553/jmcp.2024.30.12.1355.
- [2] Canadian Institute for Health Information. Prescribed drug spending in Canada, 2023: A focus on public drug programs. Technical report, Canadian Institute for Health Information, Ottawa, 2023. URL <https://www.cihi.ca/en/prescribed-drug-spending-in-canada-2023>.

- [3] Quinn Grundy, Alexandra Quanbury, Sohaib Asif Chaudhry, Alexander Hart, Farnaz Tavangar, Joel Lexchin, Marc-André Gagnon, and Mina Tadrous. Prevalence and nature of manufacturer-sponsored patient support programs for prescription drugs in Canada: a cross-sectional study. *CMAJ*, 195(46):E1565–E1576, 2023. doi: 10.1503/cmaj.230841.
- [4] Wei-Hsuan Lo-Ciganic, Julie M. Donohue, Joshua M. Thorpe, Subashan Perera, Carolyn T. Thorpe, Zachary A. Marcum, and Walid F. Gellad. Using machine learning to examine medication adherence thresholds and risk of hospitalization. *Medical Care*, 53(8):720–728, 2015. doi: 10.1097/MLR.0000000000000394.
- [5] Jianfu Yao, Harry Hochheiser, Wonjin Yoon, Eli Goldner, and Guergana Savova. Overview of the 2024 shared task on chemotherapy treatment timeline extraction. In *Proceedings of the 6th Clinical Natural Language Processing Workshop*, pages 557–569. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.clinicalnlp-1.53.
- [6] Aaron A. Tierney, Gregg Gayre, Brian Hoberman, Britt Mattern, Manuel Ballesca, Patricia Kipnis, Vincent Liu, and Kristine Lee. Ambient artificial intelligence scribes to alleviate the burden of clinical documentation. *NEJM Catalyst Innovations in Care Delivery*, 5(3), 2024. doi: 10.1056/CAT.23.0404.
- [7] Lisa Rotenstein, A. Jay Holmgren, Robert Thombley, Aditi Sriram, Reema H. Dbouk, Melissa Jost, et al. Changes in clinician time expenditure and visit quantity with adoption of artificial intelligence-powered scribes: A multisite study. *JAMA*, 2026. doi: 10.1001/jama.2026.2253.
- [8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- [9] Adam Moldwin, Dina Demner-Fushman, and Travis R. Goodwin. Empirical findings on the role of structured data, unstructured data, and their combination for automatic clinical phenotyping. *AMIA Joint Summits on Translational Science Proceedings*, 2021:445–454, 2021.
- [10] Jayetri Bardhan, Anthony Colas, Kirk Roberts, and Daisy Zhe Wang. DrugEHRQA: A question answering dataset on structured and unstructured electronic health records for medicine related queries. In *Proceedings of the 13th Conference on Language Resources and Evaluation (LREC)*, pages 1083–1097, 2022.
- [11] Gyubok Lee, Elea Bach, Eric Yang, Tom Pollard, Alistair Johnson, Edward Choi, Yugang Jia, and Jong Ha Lee. FHIR-AgentBench: Benchmarking LLM agents for realistic interoperable EHR question answering. *arXiv preprint arXiv:2509.19319*, 2025. Presented at Machine Learning for Healthcare (ML4H) 2025.
- [12] Paul Schmiedmayer, Adrit Rao, Philipp Zagar, Lauren Aalami, Vishnu Ravi, Aydin Zahedivash, Dong-han Yao, Arash Fereydooni, and Oliver Aalami. LLMonFHIR: A physician-validated, large language model-based mobile application for querying patient electronic health data. *JACC: Advances*, 4(6 Pt 1):101780, 2025. doi: 10.1016/j.jacadv.2025.101780.
- [13] Iván López, Akshay Swaminathan, Karthik Vedula, Sanjana Narayanan, Fateme Nateghi Haredasht, Stephen P. Ma, April S. Liang, Steven Tate, Manoj Maddali, Robert Joseph Gallo, Nigam H. Shah, and Jonathan H. Chen. Clinical entity augmented

- retrieval for clinical information extraction. *npj Digital Medicine*, 8(1):45, 2025. doi: 10.1038/s41746-024-01377-1.
- [14] Jinhao Jiang, Kun Zhou, Zican Dong, Keming Ye, Wayne Xin Zhao, and Ji-Rong Wen. Struct-GPT: A general framework for large language model to reason over structured data. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9237–9251. Association for Computational Linguistics, 2023.
 - [15] Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6233–6251. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.372.
 - [16] Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R. Dalal, Jennifer L. Kim, Michael Moor, Robyn Fong, Curran Phillips, Kevin Alexander, Euan Ashley, Jack Boyd, Kathleen Boyd, Karen Hirsch, Curt Langlotz, Rita Lee, Joanna Melia, Joanna Nelson, Karim Sallam, Stacey Tullis, Melissa Ann Vogelsong, John Patrick Cunningham, and William Hiesinger. Almanac — retrieval-augmented language models for clinical medicine. *NEJM AI*, 1(2):AIoa2300068, 2024. doi: 10.1056/AIoa2300068.
 - [17] Junde Wu, Jiayuan Zhu, Yunli Qi, Jingyi Chen, Min Xu, Filippo Menolascina, and Vicente Grau. Medical graph RAG: Towards safe medical large language model via graph retrieval-augmented generation. *arXiv preprint arXiv:2408.04187*, 2024. Also ACL 2025.
 - [18] Karthik Soman, Peter W. Rose, John H. Morris, Rabia E. Akbas, Brett Smith, Braian Peetoom, Catalina Villouta-Reyes, Gabriel Ceron, Yongmei Shi, Angela Rizk-Jackson, Sharat Israni, Charlotte A. Nelson, Sui Huang, and Sergio E. Baranzini. Biomedical knowledge graph-optimized prompt generation for large language models. *Bioinformatics*, 40(9):btac560, 2024. doi: 10.1093/bioinformatics/btac560.
 - [19] Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Moninder Agrawal, Xiaoyi Jiang, and David Sontag. TabLLM: Few-shot classification of tabular data with large language models. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 206 of *Proceedings of Machine Learning Research*, pages 5549–5581, 2023.
 - [20] Sung A. Lee, Siddharth Jain, Alexandr Chen, Kohta Ono, Arpita Biswas, Akos Rudas, Jason Fang, and Jeffrey N. Chiang. Clinical decision support using pseudo-notes from multiple streams of EHR data. *npj Digital Medicine*, 8(1):394, 2025. doi: 10.1038/s41746-025-01777-x.
 - [21] Moritz Engelke, Giulia Baldini, Jens Kleesiek, Felix Nensa, and Amin Dada. FHIR-Former: enhancing clinical predictions through Fast Healthcare Interoperability Resources and large language models. *Journal of the American Medical Informatics Association*, 32(12):1793–1801, 2025. doi: 10.1093/jamia/ocaf165.
 - [22] John Grimes, Ryan Brush, Nikolai Ryzhikov, Piotr Szul, Joshua Mandel, Daniel Gottlieb, Grahame Grieve, Bashir Sadjad, and Aditi Sanyal. SQL on FHIR — tabular views of FHIR data using FHIRPath. *npj Digital Medicine*, 8(1):342, 2025. doi: 10.1038/s41746-025-01708-w.
 - [23] John Grimes, Piotr Szul, Alejandro Metke-Jimenez, Michael Lawley, and Kayvan Loi. Pathling: analytics on FHIR. *Journal of Biomedical Semantics*, 13(1):23, 2022. doi: 10.1186/s13326-022-00277-1.

- [24] Yikuan Li, Hanyin Wang, Halid Ziya Yerebakan, Yoshihisa Shinagawa, and Yuan Luo. FHIR-GPT enhances health interoperability with large language models. *NEJM AI*, 2024. doi: 10.1056/AIcs2300301.
- [25] Ahmad Idrissi-Yaghir, Kamyar Arzideh, Henning Schäfer, Bahadır Eryilmaz, Maximilian Bahn, Yufan Wen, Kevin Borys, Eva Hartmann, Christoph Schmidt, Obioma Pelka, Johannes Haubold, Christoph M. Friedrich, Felix Nensa, and René Hosch. Using a diverse test suite to assess large language models on Fast Health Care Interoperability Resources knowledge: Comparative analysis. *Journal of Medical Internet Research*, 27:e73540, 2025. doi: 10.2196/73540.
- [26] Sara Kothari and Ayush Gupta. Question answering on patient medical records with private fine-tuned LLMs. *arXiv preprint arXiv:2501.13687*, 2025.
- [27] Chenhan Yuan, Qianqian Xie, and Sophia Ananiadou. Zero-shot temporal relation extraction with ChatGPT. In *Proceedings of the 22nd Workshop on Biomedical Natural Language Processing and BioNLP Shared Tasks*, pages 92–102, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.bionlp-1.7.
- [28] Scott L. Fleming, Alejandro Lozano, William J. Haberkorn, Jenelle A. Jindal, Eduardo Pontes Reis, Rahul Thapa, Louis Blankemeier, Julian Z. Genkins, Ethan Steinberg, Ashwin Nayak, et al. Medalign: A clinician-generated dataset for instruction following with electronic medical records. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 22021–22030, 2024. doi: 10.1609/aaai.v38i20.30205. URL <https://arxiv.org/abs/2308.14089>.
- [29] Hejie Cui et al. Timer: temporal instruction modeling and evaluation for longitudinal clinical records. *npj Digital Medicine*, 2025. doi: 10.1038/s41746-025-01965-9. URL <https://arxiv.org/abs/2503.04176>. Also presented at ICLR 2025.
- [30] Leah L. Zullig, Shelley A. Jazowski, Tracy Y. Wang, Anne Hellkamp, Daniel Wojdyla, Laine Thomas, Lisa Egbonu-Davis, Anne Beal, and Hayden B. Bosworth. Novel application of approaches to predicting medication adherence using medical claims data. *Health Services Research*, 54(6):1255–1262, 2019. doi: 10.1111/1475-6773.13200.
- [31] Fateme Nateghi Haredasht, Ivan Lopez, Steven Tate, Pooya Ashtari, Min Min Chan, Deepali Kulkarni, Chwen-Yuen Angie Chen, Maithri Vangala, Kira Griffith, Bryan Bunning, Adam S. Miner, Tina Hernandez-Boussard, Keith Humphreys, Anna Lembke, L. Alexander Vance, and Jonathan H. Chen. Predicting treatment retention in medication for opioid use disorder: A machine learning approach using NLP and LLM-derived clinical features. *Journal of the American Medical Informatics Association*, 32(12):1865–1876, 2025. doi: 10.1093/jamia/ocaf157.
- [32] David Anderson, Michaela Anderson, Margret Bjarnadottir, Stephen Mahar, and Shriyan Reyya. Paging Dr. GPT: Extracting information from clinical notes to enhance patient predictions, 2025. Preprint; not yet peer-reviewed.
- [33] Yanjun Gao, Skatje Myers, Shan Chen, Dmitriy Dligach, Timothy A. Miller, Danielle Bitterman, Matthew Churpek, and Majid Afshar. When raw data prevails: Are large language model embeddings effective in numerical data representation for medical machine learning applications? In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5414–5428. Association for Computational Linguistics, 2024.

- [34] Jason Walonoski, Mark Kramer, Joseph Nichols, Andre Quina, Chris Moesel, Dylan Hall, Carlton Duffett, Kudakwashe Dube, Thomas Gallagher, and Scott McLachlan. Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association*, 25(3): 230–238, 2018. doi: 10.1093/jamia/ocx079.
- [35] Christoph U. Correll, Robert E. Litman, Yuriy Filts, Jordi Llaudó, Dieter Naber, Ferran Torres, Javier Martínez, Srihari Gopal, David Hough, Adam Savitz, and Jennifer Kern Sliwa. Three-year outcomes of 6-month paliperidone palmitate in adults with schizophrenia: an open-label extension study of a randomized clinical trial. *JAMA Psychiatry*, 81(11):1099–1108, 2024. doi: 10.1001/jamapsychiatry.2024.1994.
- [36] Robert Hodges, Kristen Tokunaga, and Joseph LeGrand. A novel method to create realistic synthetic medication data. *JAMIA Open*, 6(3):ooad052, 2023. doi: 10.1093/jamiaopen/ooad052.
- [37] Daniella Meeker, Crystal Kallem, Yan Heras, Stephanie Garcia, and Casey Thompson. Case report: evaluation of an open-source synthetic data platform for simulation studies. *JAMIA Open*, 5(3):oac067, 2022. doi: 10.1093/jamiaopen/oac067.
- [38] Qiao Jin, Won Kim, Qingyu Chen, Donald C. Comeau, Lana Yeganova, W. John Wilbur, and Zhiyong Lu. MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval. *Bioinformatics*, 39(11):btad651, 2023. doi: 10.1093/bioinformatics/btad651.